

Katie Olsen

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EDUCATION

Bachelor of Science, Computer Engineering

Brigham Young University • Provo, UT • 2026 • 3.88

EXPERIENCE

Engineering Intern

Browns Hill Engineering and Controls

May 2024 -August 2024, Littleton, CO

- Programmed a Software Development Kit (SDK) that interacts with CloudFlare's SSL Certificate Bundling/Signing API.
- Created a script package and Ignition project for an employee-utilized User Interface which allows them to sign, revoke, and store certificates, as well as generate certificate-signing requests.
- Worked with Docker, Python, Ignition, MySQL, GitHub, and CloudFlare's API to take the project from start to finish.

Engineering Intern

Browns Hill Engineering and Controls

May 2023 -August 2023, Littleton, CO

- Programmed the front and back-end for an Ignition Human-Machine-Interface (HMI) on 4 projects to transfer data tags, edit tag data, and refurbish the front end FactoryTalk HMI to a visually-modernized Ignition HMI.
- Worked with another Engineer to develop a company vault which reduced duplicate information collected by field engineers and IT personnel, by merging all information into a MySQL table, and visualizing information in an Ignition User Interface.
- Created 8-9 database integration scripts and 7-8 testing scripts during the course of the internship.

Engineering Intern

Browns Hill Engineering and Controls

May 2021 - September 2021, Littleton, CO

- Developed processes for cybersecurity and troubleshooting SCADA systems and radios.
- Developed Python scripts and front/back-end Ignition HMI to read data tag files on CSV files and convert them to an Ignition-readable JSON format.
- Performed at a level sufficient to be invited back for summer of 2023.

INVOLVEMENT

Automated Pit Exploration System Capstone Project

BYU Capstone Team Project for Idaho National Laboratory (INL) • Programmer

September 2025 – April 2026

- Worked with INL in a team of five students to develop a User Interface that controls and visualizes robotic movement, with the ability to connect to two or more robots.
- Developed a pipeline between ROS2 and Unity: moving controls for a robotic model in Unity sends ROS2 commands to a physical robot, which in return sends its updated position to be visualized in Unity.
- Added functionality to connect Unity to a ZED camera and display its visual data in the Unity environment.

Brigham Young University – Antweight Battle Bot Competition

Co-team leader • Programmer

December 2023

- Programmed and designed the electronics—using an ESP32—for an Antweight competition-ready Battle Bot.
- Co-leader of a team of 4 – Bot fulfilled the requirements of less than 1 lb. and competed in the December Battle Bot competition.

Church of Jesus Christ of Latter-Day Saints

Full Time Missionary • Teaching and Leadership Experience

September 2021 - March 2023

- Taught several 30-minute gospel lessons to youth and adults daily.
- Planned and taught weekly English classes to native Spanish speakers.
- Participated in biweekly community service activities (Soup Kitchens, Park Cleanup, etc.)

Lockheed Martin - 2020 Code Quest Competition

Programmer

April 2019

- Worked 2.5 hours to solve a list of 20-30 coding challenges containing varying difficulty levels. Points awarded based on difficulty level.
- Coded in a team of three using Java to complete the challenges.

SKILLS

Front End: HTML, CSS, JavaScript, TypeScript, AWS

Back End: C++, C, C#, Python, Java, MySQL, NoSQL, Linux, Docker, RISCv Assembly, System Verilog

Hardware: Microcontrollers-Arduino, FPGA, PYNQ, Small Circuit Board design, ESP32, ROS2

Interfaces: Git/GitHub, HMI—Ignition/FactoryTalk, Virtual Machines, Unity

Languages: English (Native), Spanish (Advanced Fluency)